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Missing values imputation using Fuzzy K-Top Matching Value

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ABSTRACT

Missing data occurs when variables or observations are missing. Researchers exclude or impute influenced variables and data. This study proposes Fuzzy K-Top Matching Value (FKTM) for missing value imputation. It imputes missing numerical and categorical data with intelligent estimates based on similar records, decreasing bias. Expectation-maximization is used, where it employs fuzzy clustering to find a group of similar data and estimate them. We compare FKTM with original datasets on Immunotherapy and Cryotherapy. Multiple classification techniques are used on the imputed datasets. Random Forest achieved the best, with 93.3% for cryotherapy and 85.6% for Immunotherapy. The proposed approach is compared with Multivariate Imputation by Chained Equations (MICE) utilizing a Support Vector Machine. The proposed approach beats MICE with 82.2% accuracy. On the Cryotherapy dataset, the proposed approach surpasses existing strategies with 86.6% accuracy. Levene and Shapiro-Wilk were used to examine the homoscedasticity and normality of data after imputation. The proposed imputation procedure has no detrimental influence on the dataset. Finally, execution time and RMSE of imputed values are determined for three datasets with varied sample sizes and data dimensions. The proposed system exhibits a fast execution time and low RMSE. The proposed FKTM works well in experiments and looks promising.

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1. Introduction

Missing data might take the form of missing values or variables, both of which are common issues in economics (Zhang, 2016). When a variable is partially absent, the needed information for certain units is not collected. Non-response, which is common in any survey and can be severe, is the most common source of missing values in surveys. Non-response might mean refusing to answer any questions at all (unit non-response) or refusing to answer particular questions (question non-response) (item non-response) (Nikfalazar et al., 2020). The rates of both forms of non-response vary significantly depending on the survey. Because the essential questions were not included in a survey or the information is secret, entire variables may be missing (Khan and Hoque, 2020).

Typically, surveys are created for a specific purpose and do not include variables that are not essential. When there is missing data, one frequent approach is to study the entire dataset. When whole variables are absent from the data, the variables are omitted from the model. When data is missing, the analysis is often done on full instances, units for which all relevant variables are accessible. Another typical approach to missing data is to generate a "full" dataset by assigning values to the observations that are missing (Bertsimas et al., 2017).

Some ways assign numerous values per observation or avoid assigning explicit values altogether. Still, we use "imputation" to refer to any methods that seek to solve the problem of missing data, even though it generally refers to methods that assign explicit values. Most surveys already include imputed values for units that did not reply. These values are often anticipated based on the responses, which is impossible if the variables are entirely lacking. However, if the missing variables are found in another dataset, the full dataset can be imputed. The key distinction between partially missing variables and imputations is that in the latter situation, the source data for the imputations comes from a different data collection or era (Mittag, 2013).

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Soft computing is seen as the polar opposite of the current computing paradigm, which may be represented using words like hard, crisp, stiff, inflexible, and foolish. Soft computing may be an attempt to mimic the soft, adaptive, and intelligent nature of natural objects like animals, plants, and humans. It comprises several concepts and strategies to overcome obstacles that arise in real-world circumstances (Alamoodi et al., 2021). Our reality appears to be imprecise, ambiguous, and difficult to describe, which causes several issues. For example, the uncertainty in a measured amount stems from intrinsic fluctuations in the measuring method. Because of the external or internal impacts of these devices that gather data from the source, the value is overlooked. There are various techniques for dealing with missing values in data, but none is better than the others (Bystrov, 2000).

The only solution to the missing value problem that is genuinely effective is to remove all missing values from the dataset (Salgado et al., 2016). As a result, various situations necessitate different responses. Furthermore, the missing mechanism determines the efficacy of missing data treatment strategies. As a consequence, when the variables X and Y are supplied. There are three sorts of missing values: (1) Missing Completely at Random (MCAR): response probability is dependent on X but not on Y, (2) Missing at Random (MAR): response probability is independent of X and Y, and (3) Non-ignorable missing data is the most difficult problem to solve, but it is also the most likely to occur (Little and Rubin, 2019). MCAR data has a far higher amount of unpredictability than MAR data. Non-ignorable: The likelihood of a response is influenced by factors X and perhaps Y. In reality, meeting the MCAR assumption is frequently challenging. The assumption of MAR is used in most missing data approaches.

Imputation approaches for missing values can be classed as parametric or non-parametric based on the data structure. The parametric imputation approach is favored if users can dependably offer the dataset's parametric forms. If consumers do not know a dataset's true distribution, a non-parametric imputation is a good option. Non-parametric imputation, designed for scenarios with high sample numbers, can produce superior fits by capturing structure in the dataset. Non-parametric imputations, on the other hand, are also very common in high dimensions. The curse of dimensionality is very common, and non-parametric fitting approaches in small sample settings may fit anomalies if the data are too near (Hoti and Holmström, 2004).

There are several approaches to dealing with missing values in datasets. There are four different types of methods: (1) *Discarding missing values* when avoiding missing data is a direct solution to the problem. It has a thorough case analysis as well as an available-case analysis. (2) *Single value imputation* that includes simple missing value imputation using an average, min, max, most frequent, . . .etc., and single value random imputation, which performs attributions all the more formally. (3) *Multiple value imputation* in which the dataset is thought of as a multivariate result, any parts of which can be absent. (4) *Model-based imputation* in which missing data can be dealt with in Bugs by demonstrating the information factors that have missingness (Salgado et al., 2016).

Currently, methods based on machine learning for estimating missing values are decision trees (Ross Quinlan et al., 1994), neural networks (Zhang et al., 2003), lazy trees (Friedman et al., 1997), and dynamic path generation (White, 1987). However, because these approaches may disrupt the dataset's original distribution during the imputing process, they are not adequate solutions to missing value concerns. Furthermore, certain machine learning algorithms are limited to discrete numbers. The imputation outcome of these methods may disturb the dataset's original distribution since continuous characteristics are discretized before being handled.

In many circumstances, increasing accuracy and certainty requires significant time and money. Soft computing may be thought of as a foundation for the rapidly growing field of computational intelligence (CI). The key characteristic of computational intelligence is being based on soft computing. Hence, it includes cognitive computing, which does not derive from classical theories such as genetic algorithms (GA), neural networks (NN), and fuzzy logic (FL) (Rajasekaran and Pai, 2003). When it comes to learning and curve fitting, fuzzy logic, probabilistic reasoning, and categorization are mostly related to neurocomputing approximate reasoning. A fuzzy logic system (FLS) forms rules based on heuristic knowledge and adapts them using sample data for a wide range of real-world problems (Silva-Ramirez and Cabrera-Sánchez, 2021).

In this paper, we intend to use soft computing, especially fuzzy systems, to estimate the missing mechanisms of MAR and MCAR in a given dataset. The proposed system assumes the same circumstances of the external and internal environment during the data acquisition procedure. During the normal procedure, something goes wrong, leading to missing values. Good outcomes have been reached in several circumstances by integrating diverse soft computing strategies. The proposed system combines both fuzzy expectation maximization for global search and k-Nearest Neighbours (kNN) for local search to find Fuzzy K-Top Matching Value (FKTM). This paper's contributions are summarized as follows:

- A novel methodology for missing value imputation, FKTM, is suggested. Filling up numerical and categorical missing values with intelligent guesses that rely on records comparable to the missing value record decreases the biased guess.
- It identifies a set of comparable data and derives an estimate from it. Using a fuzzy expectation–maximization algorithm and fuzzy clustering approach. Then kNN is used to find the nearest matching value.
- Using two freely available natural datasets, Immunotherapy and Cryotherapy, we compare the performance of FKTM to that of the original datasets. For classification, Random Forests (RF), Logistic Regression (LR), and Support Vector Machine (SVM) are utilized, with RF achieving higher results. It achieves an accuracy of 93.3% in cryotherapy and 85.6% in Immunotherapy.
- The suggested approach is also compared to a well-known imputation method called Multivariate Imputation by Chained Equations (MICE), which employs the SVM classifier. In the Immunotherapy dataset, the suggested technique outperforms MICE with an accuracy of 82.2%.
- The proposed technique's accuracy is compared to five imputation techniques on the Cryotherapy dataset and outperformed all of them.
- The proposed technique is analyzed using Levene and Shapiro-Wilk statistical tests and showed that they didn't expose a negative impact on the data.
- The proposed algorithm execution time and imputation error are tested against three datasets of different sizes and showed efficient execution time and decreased error.

The rest of the paper is laid out as follows. The preliminaries are introduced in section 2. The related approaches and methods for missing value imputation are presented in section 3. The proposed system is discussed in section 4, including the foundation theory, the FKTM algorithm, and an illustrative example. Section 5 examines the proposed system performance in Immunotherapy and Cryotherapy and compares the proposed system performance with MICE. Finally, we conclude the study in section 6.

2. Preliminaries

Fuzzy set theory has a long and illustrious history. Aristotle proposed the "Laws of Thought" to build a basic logic theory and, later, mathematics. The "Law of the Excluded Middle," for example, asserts that any proposition must be True (T) or False (F). Łukasiewicz (Breuer, 1972) was the first to suggest a third zone (beyond T and F). A numeric number between T and F for the third value is provided, which may be interpreted as "possible." In his landmark book "Fuzzy Sets," Zadeh established the mathematics of fuzzy set theory, and by extension fuzzy logic, by researching four-valued logic, five-valued logic, and eventually an infinite-valued logic. According to this idea, the membership function (or the values F and T) should be extended to the real-number range [0, 1]. Fuzzy membership functions express the similarity of objects to ambiguous qualities. The membership function includes all the information that a fuzzy set conveys (Dubois and Prade, 2012).

Fuzzy systems have two major qualities that help them perform better in certain situations. Fuzzy systems are well suited to uncertain or imprecise reasoning, particularly for systems with difficult-to-deduce mathematical models. Under incomplete or unclear knowledge, decision-making using estimated values is also possible with fuzzy logic. Gray or ambiguous information is treated in science as black-and-white mathematical facts. Humans and fuzzy systems can use heuristic linguistic rules when there is no mathematical model for the problem (Alonso Moral et al., 2021).

FL merely calculates the fuzzy similarity of instances and concludes (Fahiman et al., 2017). They can function efficiently and successfully in noisy environments and deal with missing data. As a result, they do not require initial estimations for missing data. They can also cope with erroneous data with ease. The missing value imputation domain requires approaches that can quickly and efficiently deal with imprecise data. Fuzzy approaches are suited for the imputation of missing data for the reasons stated (Amiri and Jensen, 2016).

The essential principle of the well-known EM technique is used in Expectation-Maximization Imputation (EMI). The expectation step (E-Step) and the maximization step (M-Step) are the two key phases in the EM algorithm (Junninen et al., 2004). Before imputing missing values, the mean and covariance values are calculated using the E-step technique. Therefore, these values are utilized to impute the missing data. The maximum likelihood approach is used to get the best mean and covariance values based on the outcoming data. Therefore, mean and covariance values are modified using M-step to enhance the imputation values. The operation can be repeated to increase imputation quality (Rahman et al., 2016).

3. Literature review

Missing value handling has seen the greatest research and usage of machine learning algorithms and methods derived from statistical learning theory. Machine learning-based imputation methods are sophisticated approaches that involve employing unsupervised or supervised learning to construct a prediction system to manage missing data. Like other imputation methods, these approaches rely on information from non-missing values in the data to estimate missing data estimations. An imputation with high prediction accuracy may be maintained if the provided data offers significant information for resolving missing values (Emmanuel et al., 2021). This section contains related research including the most recent findings on the subject of missing value imputation.

This is a hot research topic, and many studies discuss the issue. For example, Li et al. (2021) suggested two algorithms: jointly fuzzy C-Means and VQNN (Vaguely Quantified Nearest Neighbor) imputation (JFCM-VQNNI) and jointly fuzzy C-Means and fitted VQNN imputation (JFCM-FVQNNI), by taking into account clustering principles and adequate extraction of ambiguous data in JFCM-VQNNI and JFCM-FVQNNI algorithms, the missing value is handled as a decision feature, and predictions are supplied for objects with at least one missing value. The JFCM-VQNNI method finds the likely nearest filled values by comparing comparable items and clusters and using indistinguishable matrices, tolerance relations, and fuzzy membership relations. The JFCM-FVQNNI algorithm synthetically examines the fuzzy membership of the dependent features for instances in each cluster using the JFCM-VQNNI approach. By assessing highly relevant decision characteristics, the JFCM-FVQNNI technique executes fuzzy decision membership modification in each item about the connected clusters to fill the missing values more precisely. The recommended JFCM-FVQNNI and JFCM-VQNNI algorithms provide acceptable and tolerable imputation performance on five datasets, according to the testing.

Karmitsa et al. (2022) suggested a method for missing value imputation that combines two well-known techniques based on clusterwise linear regression: linear regression and clustering. They used only data points compared to incomplete data points to estimate missing values. Clustering-based imputation approaches employ a similar concept. Nonetheless, the linear regression technique is employed inside each cluster to properly forecast missing values, which is done concurrently with clustering. The suggested technique is validated using synthetic and real-world datasets, demonstrating good results in imputations in MCAR and MAR datasets with a clear structure and missing data percentages of less than 25%.

Huang et al. (2020) estimated missing values in datasets by creating a GA-based integrated FCM imputation technique. The GA was utilized to enhance two key parameters in the FCM model: membership degree and cluster centroids. The FCM used a clustering approach to handle the missing variables. A matrix-based data structure was adopted to depict better data similarity within the specific context of the integrated imputation technique. The efficiency of the integrated imputation technique is demonstrated by an experimental evaluation using taxi Global Positioning System (GPS) data in Manhattan, New York City. The average accuracy derived by the integrated imputation approach under the ±5% and ±10% criteria is 0.576 and 0.785, respectively. The Euclidean distance, on the other hand, performs better than the Manhattan distance.

Razavi-Far et al. (2020) proposed that kEMI and kEMI+ are two approaches for missing data imputation based on the kNN algorithm for pre-imputation and the Expectation-Maximization algorithm for posterior-imputation, respectively. The former is a local search technique that tries to figure out the ideal value for k automatically. In contrast, the latter learns global similarities from the best k-NN to mimic missing scores. kEMI+ uses an information fusion approach to combine top estimations using the Dempster–Shafer fusion module to arrive at the final estimate. They are capable of working with both numerical and category information. The suggested missing data imputation algorithms are tested on twenty-one publically available datasets with varying missingness and ratios, and the findings show that they are effective.

Rantou et al. (2017) considered imputation methods on structural time series models, state-space representation of an Arima model and Kalman Smoothing, linear and spline interpolation, Exponentially weighted moving average (EWMA), and the overall mean are all used. Using the statistical program R, they examined and measured the performance of imputation procedures in the setting of univariate time-series data. The imputation strategies are contrasted using two error metrics, MRSE and MAPE, and three main types of time series patterns are addressed. They based the

experiment on two datasets, bank of America data and France Flu data, with a missingness rate of 0.8. The experiment results showed that structural models using Kalman smoothing and linear interpolation could perform better than other algorithms when dealing with values data in univariate time series.

Morid et al. (2017) produced similarity-based classification models with adequate accuracy increases by looking into effective temporal feature engineering and reduction, missing value imputation, and change-point detection methodologies. They used a piecewise aggregation approximation strategy to extract fine-grain temporal characteristics and proposed a simple mechanism to impute missing values in temporal features. According to their findings, the k-NN technique outperforms the relevant benchmarks for predicting early intensive care unit (ICU) death. By developing and improving temporal feature engineering and reduction approaches for similarity-based time series classification, they contributed to time series classification and early ICU mortality prediction. They present an adjacency-based imputation approach that substitutes missing values with their posterior values if their prior values are unavailable or with their prior values if their posterior values are unavailable. If both are known, the imputed value is the average of the previous and posterior values. Regardless of the distance function used, the proposed imputation can be done. They provide similarity-based time series classification by using the kNN algorithm with fine-grain temporal features (F), missing-value imputation (I), feature weight assignment (W), and change point detection (C) methods (kNN-TSC-FIWC). With Window size 1, the accuracy was 0.72 without change point detection and 0.80 with change point detection.

Wellenzohn et al. (2014) imputed missing values in time series data streams by creating top-k case matching (TKCM). For each time series, TKCM constructed a set of reference time series and computed the imputation using similar historical events in the reference time series. They supported non-linearly connected and phase-shifted time series. Because TKCM was responsive to successively missing values, the accuracy of the imputed values did not suffer when blocks of values were absent. They used real-world and synthetic data, including the SBR dataset of South Tyrol meteorological time series, Flights, and Chlorine databases.

Pforte et al. (2017) used the Bayesian method called Markov Chain Monte Carlo to fill in the time series gaps and generate error limits, reflecting the amount of confidence in the imputation. The whole method was written in open-source tools, such as R and JAGS, to make it accessible to other users and ensure that any results are repeatable. The experiment was based on the ESPON database. However, it offers a broad approach to missing data imputation in space–time data series as a method. As a result, it may be used to impute data from short-run time series that are also part of a spatial hierarchy.

Pratama et al. (2017) reviewed missing data handling methods from the conventional ones, such as deletion, mean, and mode imputation, to the modern ones. They also discussed some reasonable methods to estimate missing values to be used by other researchers in this missing value imputation. As missing data ensued according to several problems called missingness mechanisms, they conducted data measurements under different conditions. The missingness mechanism could be like MAR, NMAR, and MCAR.

Pampaka et al. (2016) established a procedure for dealing with missing data with multiple imputations (MI) using their dataset. The procedures started with identifying missing data, then investigating missing data patterns, which defined variables in the dataset that could be related to missing values to be used in the imputation model, then imputed missing data to produce 'm' complete datasets, then ran the models of interest using their imputed datasets, and finally combining the parameters of their models, and reporting the final model. The MI's findings are encouraging, especially given the high quantity of missing data (over 60% for dropouts) and the fact that it was missing on a 'simple' binary variable. The model tested is highly imprecise (Nagelkerke's pseudo-R^2 14 0.237 for the entire dataset).

Azur et al. (2011) introduced multivariate imputation by chained equations (MICE). Given the variables used in the imputation approach, it operates on the premise that the missing values are MAR. The MICE approach employs a series of regression models, with each variable with missing data modeled conditionally on the other variables in the data. Each variable, for example, binary variables with logistic regression and continuous variables with linear regression, may be modeled depending on its distribution. This series should be continued until convergence appears to have been achieved. This procedure is repeated until all of the variables supplied have been imputed. The final imputations are kept after these cycles, resulting in a single imputed dataset. The information density of the dataset will determine the accuracy of the imputations. If the dataset contains independent variables with no correlation, imputations will be incorrect.

Orczyk and Doroz Rafałand Porwik (2021) imputed missing data and independently employed the k-NN classifier on a single feature. The method is called (k-NNI). Unlike traditional approaches, which could only employ full vectors in the imputation process, this method's methodology permits data from incomplete vectors to be used to impute another incomplete vector. The single-dimensional subspaces enable the usage of as much data as feasible, including values from incomplete vectors. Furthermore, because each feature is processed separately, many features can be processed in parallel simultaneously. It also eliminates the need for any kind of data standardization or scaling. Table 1 summarizes the methods used in the literature review.

Imputation and missing data research have focused on the conditions under which they provide unbiased estimates. These constraints are regularly violated. The used machine learning-based imputation algorithms in the literature leverage record similarity or feature correlations (Chen et al., 2017). These machine learning-based missing data imputation approaches may be divided into two types depending on the similarity structure of the search and the imputation process (Feng et al., 2015). The first group includes missing data imputation approaches that look for global similarity within the data structure. In contrast, the second category includes techniques that look for local similarities within the data structure. Most related works focus only on one of these categories.

An FKTM is suggested as a unique missing value imputation approach for overcoming the problem. It reduces the biased guess by filling in numerical and categorical missing values with intelligent guesses based on entries similar to the one with the missing data. It uses both global similarities via expectation–maximization and local similarity via k-NN. It uses a fuzzy expectation–maximization algorithm and fuzzy clustering technique to locate a group of comparable data and create an estimate based on the group. To impute missing values, expectation–maximization uses the whole dataset and looks for correlations between the features with missing scores (target features) and the features without missing scores (covariates). Also, k-NN local similarity-based imputation looks for local similarity patterns among records to impute missing scores.

4. Proposed system

When tackling real-life classification assignments, missing or unknown data is a prevalent issue that pattern recognition systems must cope with. When it comes to data mining, one of the most difficult tasks is missing value imputation. In the preprocessing step

Table 1
Summary of literature review.

Study	Analysis	Methods	Datasets	Accuracy (%)
Li et al. (2021)	Missing value imputation in a vague environment	JFCM-VQNNI and JFCM-FVQNNI	Cleveland, Pima, Mice protein expression, Yeast, Gene expression	Mice protein expression. 100, Cleveland. 60.6 Pima. 81.8
Karmitsa et al. (2022)	Preprocessing incomplete data	Clusterwise LR	Iris plant, Wine recognition, TSPLIB1060, Red wine quality, abalone	–
Huang et al. (2020)	Integrated imputation for missing values	FCM based on GA	taxi GPS	78.5
Razavi-Far et al. (2020)	Missing data imputation	kEMI and kEMI+	14 datasets	–
Rantou et al. (2017)	Handling missing values	State Space representation, Kalman Smoothing, linear and spline interpolation, EWMA, and the overall mean	univariate time-series data	–
Morid et al. (2017)	Impute missing values in temporal features	adjacency-based imputation	Mortality dataset	82
Wellenzohn et al. (2014)	Imputation of missing values in time-series data	TKCM	4-time series datasets	–
Pforte et al. (2017)	Missing data imputation	Markov Chain Monte Carlo	ESPON database	–
Pampaka et al. (2016)	Handling missing data using multiple imputations	MI	Dropout dataset	–
Azur et al. (2011)	Multivariate imputation by chained equations	MICE	children's mental health services	–
Orczyk and Doroz Rafałand Porwik (2021)	Missing value imputation	k-NNI	WINE, WDBC, CTG, MESS, MESO, CRYO	WINE. 97.6, WDBC. 93.1, CTG. 81.4, MESS. 57.7, MESO. 97.3, CRYO. 83.8

of data mining, missing values in datasets should be extracted or estimated before being utilized for classification, association rules, or clustering. Because the aim is to impute missing data, the imputed values are the important decision variables for each model.

Our FKTM algorithm's main concept is to look for the k occurrences in the data where the measured values at a set of reference points were the most comparable to the measured values. The values at the k occurrences are then used to determine the missing value. We use the fuzzy expectation maximization technique, which effectively responds to the top-k queries by scanning, on average, a relatively small percentage of the data, as opposed to a naïve approach, which would have to scan the full data set to determine the top-k most comparable instances.

We have designed a framework to handle missing values by suggesting a suitable value; see Fig. 1. The imputed value is powered by a fuzzy system and membership manipulation. Every dataset is considered a set of instances. Every instance is acquired from data sensors. The total number of sensors is considered as the dimensions of the dataset. Different factors can affect the data values during collecting data, resulting in missing value(s). The missing values affect the regression, prediction, and classification process and can lead to the loss of the original data distribution. In turn, we have designed our framework to simulate such cases and seek a reasonable imputation methodology.

The proposed framework uses any classification dataset, with distinct instances collected under the same circumstances and assumptions. Since attributes, the columns, in every dataset are allocated once the collecting of data process is started, they are fixed and never changed over all the dataset instances. Hence, every instance is gathered under the same state that others are collected, which is the essential assumption for our proposed algo-

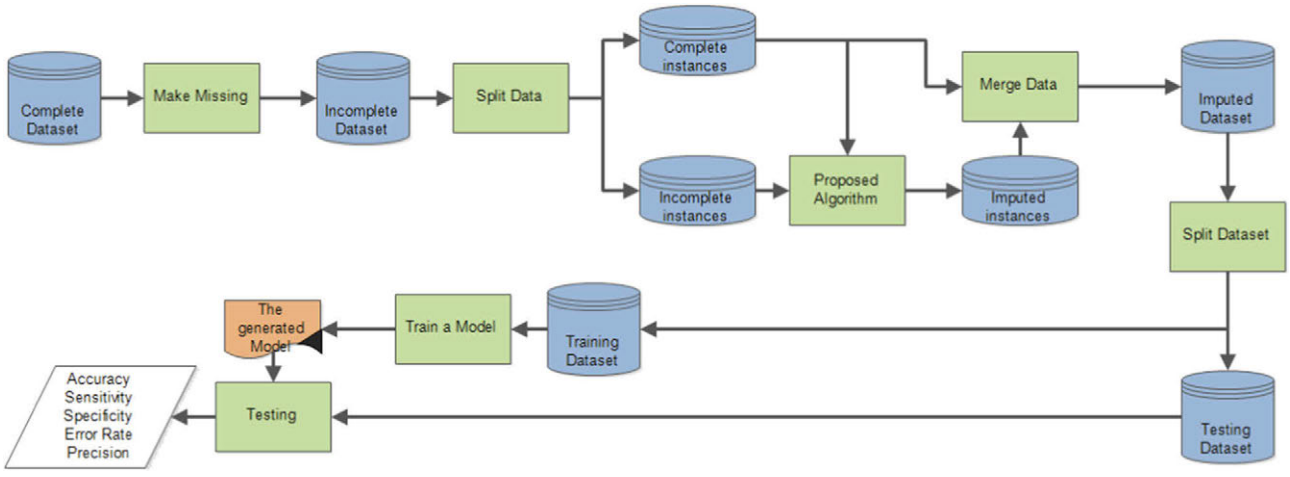


Fig. 1. The proposed framework.

rithm. Next, the framework includes a procedure responsible for randomly creating missing values in the dataset. The missed values have occurred randomly in different columns except the class/label with a given percentage. The framework divides the incomplete dataset into two sets to apply the suggested algorithm: the first set contains full instances, while the second contains incomplete instances.

The proposed algorithm accepts separated instances into two sets and returns a new set of imputed instances. The proposed algorithm is called Fuzzy K Top Matching, FKTM. It is designed to scan the set of complete instances and return the closest K complete instances. The current incomplete instance should be similar. For every missing value in an incomplete instance, FKTM creates a fuzzy set with m values and a membership of 1, where m is the total number of complete occurrences. In other words, FKTM assumes the missed values can be any value of previously occurred values incomplete instances. Next, FKTM scans the complete instances once per time and calculates the total distance between the current incomplete instance and all complete instances.

In turn, FKTM updates the fuzzy set one element per scan iteration to update the dissimilarity between the current, complete, and incomplete instances. Due to the finishing of dissimilarity calculation, FKTM sorts the elements in the fuzzy set based on the membership value in descending order. Finally, FKTM selects the same value of the top closest instance to fill the missed value in an incomplete instance. The process of FKTM repeats for every instance in the incomplete instances set to produce a new imputed set. Algorithm 1 shows the FKTM algorithm in steps. FKTM uses the Euclidian distance to measure the total distance between the complete V(C) and incomplete V(I). For instance, see equation (1).

$$distance = |V(C) - V(I)| \tag{1}$$

Based on the aforementioned, the proposed framework introduces missing values in the original dataset resulting in instances in the dataset that have been missed. These missing instances are imputed using FKTM, resulting in the imputed dataset. The data is split into training and testing sets to evaluate this system. The training set is used for ML model training, the testing set is used for testing the model, and the test predictions are utilized to get the evaluation results.

Algorithm 1: Fuzzy K-Top Matching Algorithm

Input: Dataset contains missing values
Output: Imputed dataset
topK={dummy1,. . . . . .,dummyk}, score(dummyi) = 0
kMin score = 0 // topK buffer's smallest score
for p = 1 to *N* **do**
  for all lists List(*1<= I <=M*) **do**
    Let < item-id *e*, e[*I*] > be the p-th item in *List*
    // *Score(e)* is calculated using distance.
    Score(e) = 0
      for q = 1 to M **do**
        Score(e) = $w_I e[q]$ -$w_{I-1} e[q-1]$
      end for
    //model the remaining values in PDFs.
    Update-gPDF($gPDF_i, e[I]$)
    // Topk buffer is being updated.
    if Score(e) > kMinScore then
      if e!∈ topK then
        Let z be the topK tuple with the lowest score.
        topK = topK - {*z*}
        topK = topK U {*e*}
      end if
      kMinScore = min{Score(*x*) | *x* ∈ topk}
    end if
    // Calculate confidence
    $\mu_{function} = S, (1 - \Delta)$
  end for
  return (topk, $\mu_{function}$);
end for

Next, we are discussing an example of data that has missing values and how the value would be imputed using the proposed method. Table 2 shows the data with a missing value (see Table 3 and Table 4).

From the previous table, all data is completed except row number 7 has the missing source to get this missing value. There are several steps. The first step is to compute the delta Δ between data in row 7 and all rows as following

$$\Delta_{7-1} = |15.0 - 15.0| + |16.0 - 15.9| + |14.3 - 14.1| = 0.3$$

$$\Delta_{7-2} = |15.0 - 15.2| + |16.0 - 15.7| + |14.3 - 13.9| = 0.9$$

$$\Delta_{7-3} = |15.0 - 15.2| + |16.0 - 15.8| + |14.3 - 14.1| = 0.6$$

$$\Delta_{7-4} = |15.0 - 15.0| + |16.0 - 15.9| + |14.3 - 14.2| = 0.2$$

$$\Delta_{7-5} = |15.0 - 15.3| + |16.0 - 15.7| + |14.3 - 14.5| = 0.8$$

$$\Delta_{7-6} = |15.0 - 15.2| + |16.0 - 16.0| + |14.3 - 14.1| = 0.4$$

The next step is to compute the membership function $\mu_{function}$. A fuzzy set's membership function is an extension of a classical set's indicator function. It is an extension of value in fuzzy logic that indicates the degree of truth. Degrees of truth are commonly confused with probabilities. Still, they are fundamentally different since fuzzy truth refers to membership in ill-defined sets rather than the likelihood of an event or condition occurring. Membership functions operated on the domain of all conceivable values and had a range of values ranging from 0 to 1. And it can be computed by Eq. (2).

$$\mu_{function} = \{S, (1 - \Delta)\} \tag{2}$$

The expression (1-Δ) is to represent how much it is closer to the source S

$$\mu_{function} = \{(16.1, (1 - 0.3)), (15.8, (1 - 0.9)), (15.9, (1 - 0.6)), (16.2, (1 - 0.2)), (16.5, (1 - 0.8)), (16.1, (1 - 0.4))\}$$

$$\mu_{function} = \{(16.1, 0.7), (15.8, 0.1), (15.9, (0.4), (16.2, (0.8), (16.5, (0.2), (16.1, (0.6)\}$$

From $\mu_{function}$ the missing value in row 7 is much closer to row 4, so the missing value will be 16.2.

Table 2
Example data with missing values.

U = $e_i \in W$	R_s			S
	r_1	r_2	r_3	
1	15.0	15.9	14.1	16.1
2	15.2	15.7	13.9	15.8
3	15.2	15.8	14.1	15.9
4	15.0	15.9	14.2	16.2
5	15.3	15.7	14.5	16.5
6	15.2	16.0	14.1	16.1
7	15.0	16.0	14.3	?

Table 3
The closer value to the missing value.

Δ	v	S	Top k	S
Δ_{7-1}	0.3	16.1	×	×
Δ_{7-2}	0.9	15.8	×	×
Δ_{7-3}	0.6	15.9	×	×
Δ_{7-4}	0.2	16.2	S4	16.2
Δ_{7-5}	0.8	16.5	×	×
Δ_{7-6}	0.4	16.1	×	×

Table 4
Data after imputation.

U = $e_i \in W$	R_s			S
	r_1	r_2	r_3	
1	15.0	15.9	14.1	16.1
2	15.2	15.7	13.9	15.8
3	15.2	15.8	14.1	15.9
4	15.0	15.9	14.2	16.2
5	15.3	15.7	14.5	16.5
6	15.2	16.0	14.1	16.1
7	15.0	16.0	14.3	16.2

5. Experimental results

5.1. Datasets

We used two well-known datasets from the UCI online repository to evaluate the FKTM imputation algorithm. The first dataset is Immunotherapy (Khozeimeh et al., 2017a) (Khozeimeh et al., 2017b). The second dataset is Cryotherapy (Khozeimeh et al., 2017a) (Khozeimeh et al., 2017b). The dataset contains numerical and categorical features. Missing values are created randomly in some cells and picked randomly in others. Missing values were identified as missing by setting their values to "?" at random. The two datasets have 90 instances and 7 real-valued properties. The specifics of the Immunotherapy and cryotherapy datasets are shown in Table 5 and Table 6, respectively.

We artificially introduce missing values in pure datasets, which are subsequently imputed by the provided methodologies. Because we know the original values of the purposely generated missing data, we can evaluate the strategies' performance. Imputation performance is influenced by the missing data amount (Junninen et al., 2004). There are many different types of missing data. For example, each record may have just one missing value in one case (type), while each record may have several missing values in another, although both datasets may have the same total number of missing values. Furthermore, the likelihood of a value being absent has little to do with the missing value (Rubin, 1976) (Schneider, 2001). Hence missing values are frequently random and difficult to articulate. As a result, in this experiment, we use several types of missing data, such as simple, medium, complicated, and blended.

5.2. Performance metrics

The suggested method's imputation accuracy is assessed using five well-known assessment criteria, including accuracy, sensitivity (TPR), specificity (TNR), precision (PPV), and error rate. We will now briefly define the evaluation criteria. Let N be the number of fictitious missing values, $O_i(1 \le i \le N)$ be the true value of the ith missing value that was manufactured artificially, I_i be the ith missing value's imputed value (Junninen et al., 2004). The following formulae can be used to calculate the measures:

Table 5
Immunotherapy dataset details (Khozeimeh et al., 2017a) (Khozeimeh et al., 2017b).

Feature	Attribute	Type	Values
1	Gender	Categorical	49 female and 41 male
2	Age	Numerical	15–56
3	Elapsed time before therapy	Numerical	0–12 (months)
4	Warts number	Numerical	1–19
5	Warts type	Numerical	22 plantar, 47 common, and 21 both
6	Area	Categorical	6–900 (mm^2)
7	Treatment response	Categorical	Yes/ No

Table 6
Cryotherapy dataset details (Khozeimeh et al., 2017a) (Khozeimeh et al., 2017b).

Feature	Attribute	Type	Values
1	Gender	Categorical	43 females and 47 males
2	Age	Numerical	15–67
3	Elapsed time before therapy	Numerical	0–12 (months)
4	Warts number	Numerical	1–12
5	Warts type	Categorical	9 plantar, 54 common, and 27 both
6	Area	Numerical	4–750 (mm^2)
7	Treatment response	Categorical	Yes/ No

$$\text{Accuracy} = \frac{t_p + t_n}{t_p + f_p + f_n + t_n} \tag{3}$$

$$\text{Sensitivity(TPR)} = \frac{t_p}{t_p + f_n} \tag{4}$$

$$\text{Specificity(TNR)} = \frac{t_n}{f_p + t_n} \tag{5}$$

$$\text{Precision(PPV)} = \frac{t_p}{t_p + f_p} \tag{6}$$

$$\text{Error Rate} = \frac{|ApproximateValue - ExactValue|}{|ExactValue|} \tag{7}$$

where $t_p,\ f_p,\ f_n,\ and\ t_n$ True positive, false positive, false negative, and true negative are all terms that can be used interchangeably. True positive rate, true negative rate, and positive predictive value are the acronyms TPR, TNR, and PPV, respectively. The percentage of correct predictions produced throughout the test dataset defines accuracy as a performance statistic. Sensitivity is the percentage of projected true positives to all positives in the test. The percentage of projected true negatives to all negatives in the dataset is known as specificity. The percentage of expected true positives to all predicted positives is known as precision. The error rate is the difference between the expected and actual value divided by the actual value.

The difference between values (sample or population values) predicted by a model or an estimator and the values observed is usually measured using the root-mean-square error (RMSE). The RMSE is the quadratic mean of the square root of the second sample moment of the discrepancies between expected and observed values, as shown in equation (8) (Chai and Draxler, 2014).

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{N} (\hat{y}_i - y_i)^2}{N}} \tag{8}$$

where N is the number of predicted values and $\hat{y}_i$ is the ith predicted value while y_i is the ith actual value. When computations are made outside the data sample used for the estimate, the deviations are referred to as prediction errors or residuals. We also would use RMSE to analyze our proposed approach.

Also, we use the Shapiro-Wilk (Shapiro and Wilk, 1965) statistical tests, which check the premise that the samples are drawn from a population with a normally distributed population. Another statistical test, checks the assertion that all input populations have the same variance, which is the null hypothesis. Multiple hypothesis testing (Shaffer, 1995) is also used in conjunction with the Bonferroni correction (Perneger, 1998), which divides the prespecified significance threshold (Type I error rate) by the number of hypothesis tests run. This method is notorious for over-correct, lowering the statistical power of the tests, especially when the number of hypotheses is large or the tests are strongly linked. We employed the two-sample *t*-test since the theoretical null distribution was unavailable (Keselman et al., 2004).

Hardware and software information is included in our implementation environment. The suggested solution was tested on a machine with an i7 CPU, 4 GB DDRAM, and a NIVIDA GeForce 920M graphics card. The Software environment, on the other hand, provides MATLAB 2016 x64-bit with CUDA 7 support.

5.3. Results

Three classifiers were used to evaluate the efficacy of the proposed imputation approach FKTM. The initial classifier is the SVM. SVM is a sort of supervised learning model that examines data for classification and regression analysis using learning methods. An SVM training approach builds a model that assigns new instances to one of two categories based on a collection of training examples that are individually recognized as belonging to one of two categories, making it a non-probabilistic binary linear classifier. An SVM model represents the instances as points in space mapped to separate the examples of the various categories by a significant distance.

Random forests, also known as random decision forests, are ensembles of numerous individual decision trees that work together as a classification system. It's an ensemble learning approach for classification, regression, and other tasks. It involves training a large number of decision trees and then outputting the class, that is, the mode of the classes (classification) or mean prediction (regression) of the individual trees. Random decision forests solve the problem of decision trees overfitting their training set.

Logistic regression is the final classifier. In its most basic form, logistic regression is a statistical model that uses a logistic function to describe a binary dependent variable. Logistic regression is used in regression analysis to estimate the parameters of a logistic model (a form of binary regression). A dependent variable with two alternative values, such as pass/fail, is theoretically represented by an indicator variable with the two values labeled "0" and "1" in a binary logistic model. The log-odds for the value labeled "1" in the logistic model are a linear combination of one or more independent variables ("predictors") that might be binary or continuous. Because the chance of the item labeled "1" swings between 0 and 1, the logistic function transforms log odds into probability.

The two used datasets for evaluating the proposed model have been divided into two parts training versus validating and testing, with 80% for training, 10% for validating, and the rest 10% for testing. The training dataset is the set of data used by the machine learning model to learn or to fit the parameters to the model. The validation data is a set of data used to fine-tune the model's parameters. The testing set is a collection of data that is primarily used to assess the performance of a fully specified model. To see how effective the suggested FKTM imputed approach is, the original datasets have been classified using the three classifiers SVM, Random forests, and logistic regression against the imputed datasets. The findings were calculated independently for the original and imputed datasets. The obtained results data is then compared.

5.4. Discussion

The accuracy of the original cryotherapy dataset outperformed the imputed cryotherapy dataset in the SVM classifier. On the other hand, the accuracy of the original immunotherapy dataset is less than that of the imputed dataset, which means the imputed proposed method can gain the correct value, leading to classifying the class's incorrect manner. Fig. 2 shows the comparison of the different measures, including sensitivity, specificity, and precision values, between the original and the imputed dataset based on the SVM classifier. The sensitivity of the original Cryotherapy dataset

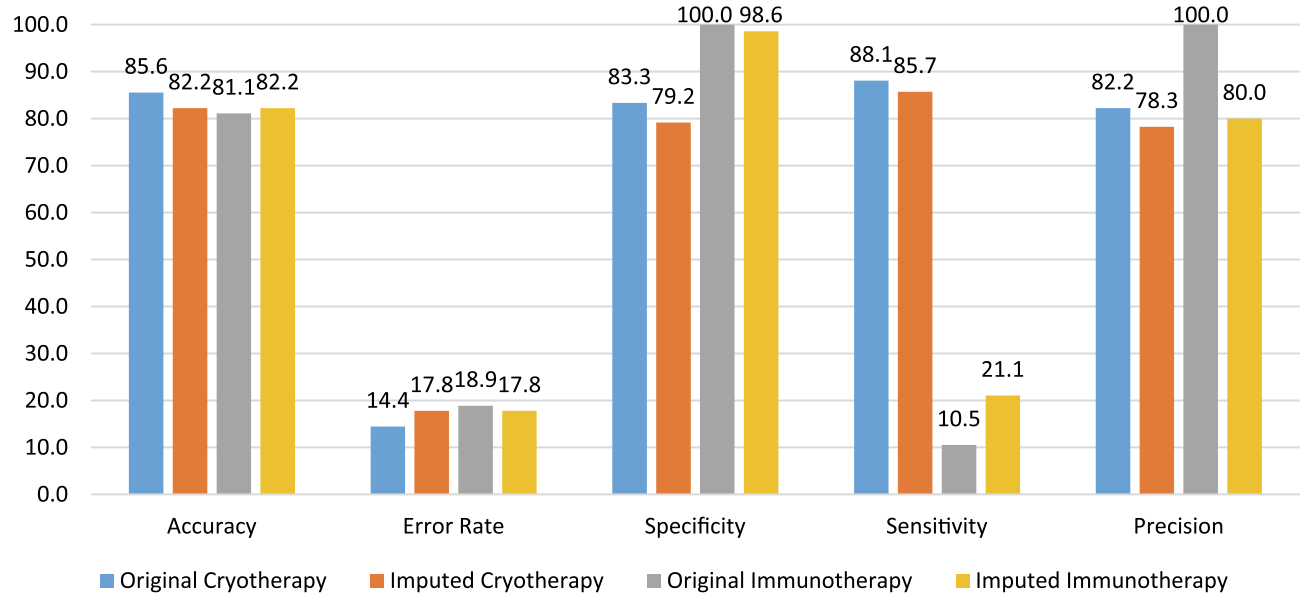


Fig. 2. Comparison of the original and the imputed dataset based on the SVM classifier.

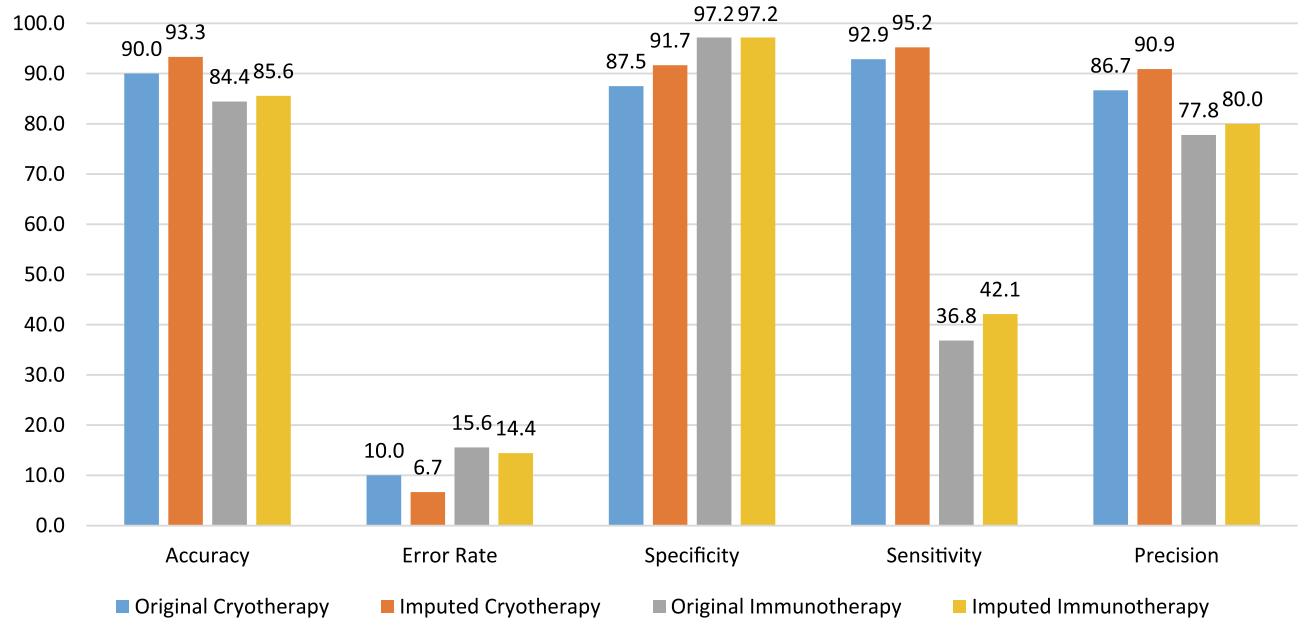


Fig. 3. Comparison of the original and the imputed dataset based on the RF classifier.

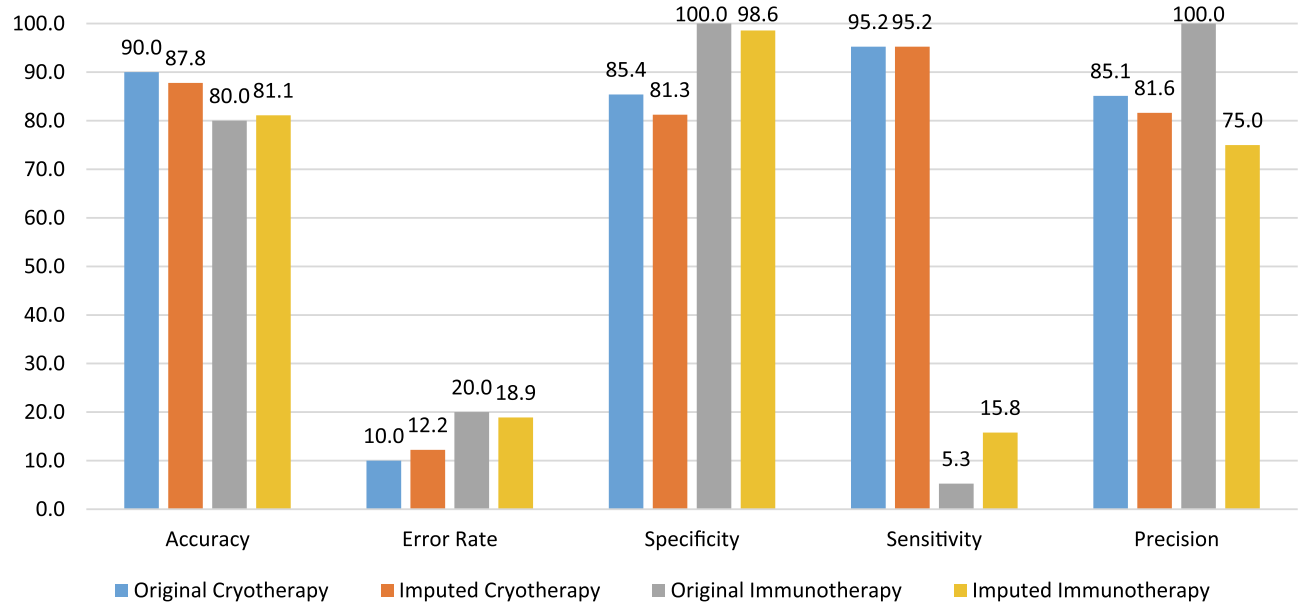


Fig. 4. Comparison of the original and the imputed dataset based on the LR classifier.

has exceeded the sensitivity of the imputed cryotherapy. The original dataset sensitivity only exceeds the imputed dataset sensitivity by 0.03, which means a tiny value. On the other hand, the sensitivity of the imputed Immunotherapy dataset has exceeded the original dataset by 0.1.

The imputed cryotherapy and immunotherapy datasets were more accurate than the original cryotherapy and immunotherapy datasets in the RF classifier. On the other hand, the sensitivity values for all imputed datasets have exceeded the value of the original datasets. The specificity was very different because the specificity of the imputed cryotherapy dataset exceeded the values of the original dataset while it was equally in the imputed and the original Immunotherapy dataset. In the same manner of accuracy, the precision of the imputed cryotherapy and immunotherapy datasets has exceeded the precision of the original cryotherapy and immunotherapy datasets. Fig. 3 shows the comparison of the different measures between the original and the imputed dataset based on the RF classifier.

A comparison has been carried out between the original and the imputed datasets based on a logistic regression classifier. The accuracy of the imputed dataset and the original dataset noted for the imputed cryotherapy dataset was less than the original one. At the same time, it exceeds imputed Immunotherapy against the original dataset. For specificity, it was noted that the imputed datasets have been failing against the original datasets. For sensitivity, the imputed and the original cryotherapy datasets were equal. The precision of the imputed cryotherapy and immunotherapy datasets was less than the original datasets. Fig. 4 shows the comparison of the different measures between the original and the imputed dataset based on the RF classifier.

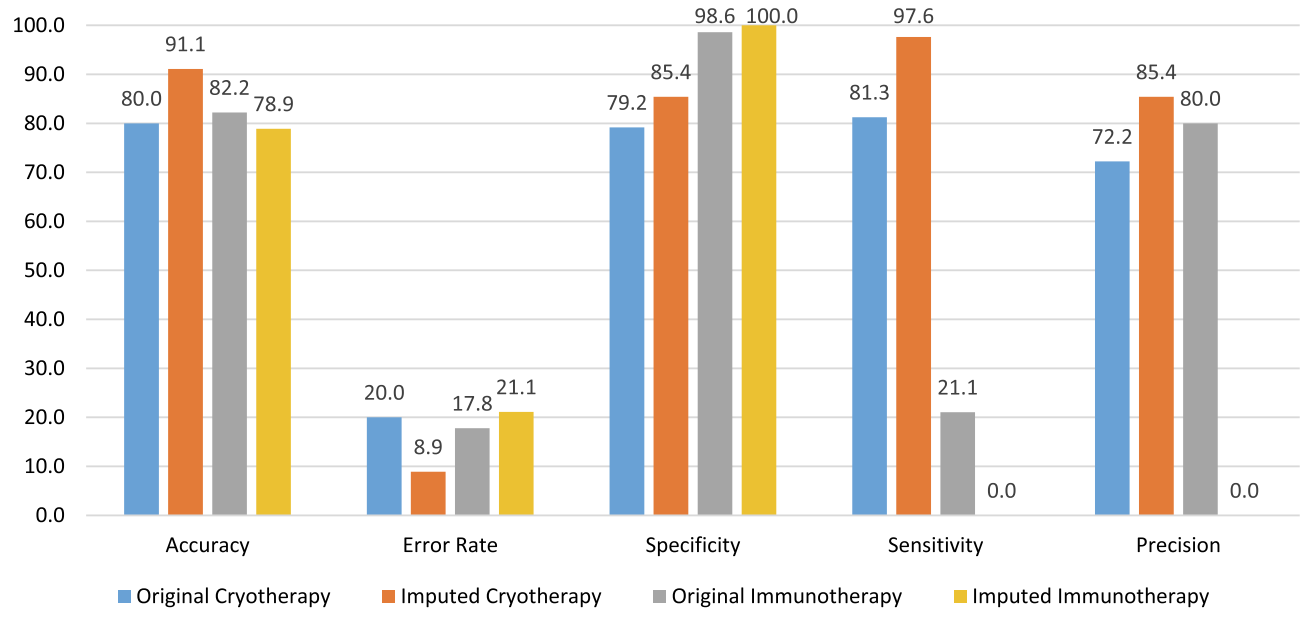


Fig. 5. Comparison of the proposed system and MICE based on the SVM classifier.

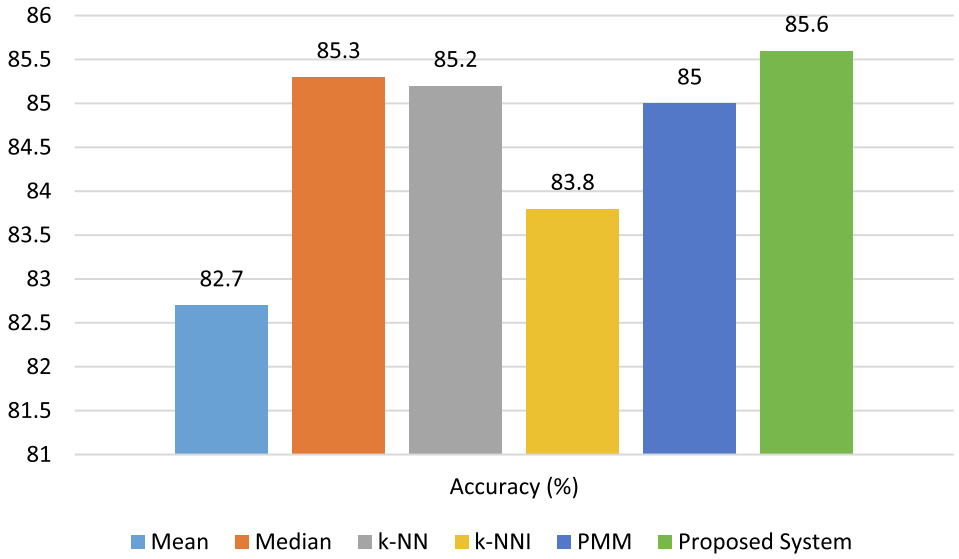


Fig. 6. Comparison between the proposed system and other imputation techniques' accuracy on the cryotherapy dataset.

Table 7
Multiple hypothesis test results comparison on cryotherapy dataset.

Method	Sex	Age	Time	Number_of_Warts	Type	Area
MICE	0.419	0.001	0.001	**0.478**	0.001	0.094
Mean	0.419	0.001	0.001	0.469	0.001	0.095
Median	0.419	0.001	0.001	**0.478**	0.001	0.098
k-NN	0.419	0.001	0.001	0.421	0.001	**0.108**
Proposed System	0.419	0.001	0.001	0.461	0.001	0.096

Comparison with the MICE (Azur et al., 2011) imputation method is made using SVM. The comparison results utilizing SVM between the proposed imputation technique and MICE are shown in Fig. 5. The suggested technique outperforms MICE in the Immunotherapy dataset, whereas, in the Cryotherapy dataset, the converse is true.

A comparison between the proposed method accuracy based on the LR classifier and other imputation techniques on the Cryotherapy dataset is shown in Fig. 6. These other imputation techniques include k-NN, mean, and median, applying the k-NN classifier to each feature independently (k-NNI), and MICE's implementation of Predictive Mean Matching (PMM) (Orczyk and Doroz Rafałand

Table 8
Multiple hypothesis test results comparison on immunotherapy dataset.

Method	Sex	Age	Time	Number_of_Warts	Type	Area
MICE	0.863	0.062	0.004	0.490	0.120	0.586
Mean	0.863	0.070	0.008	0.557	0.197	0.586
Median	0.863	0.069	0.007	0.577	0.157	0.586
k-NN	0.863	**0.072**	**0.012**	0.617	0.195	0.586
Proposed System	0.863	0.061	0.004	**0.620**	**0.415**	0.586

Table 9
Levene test (significance level of 0.10).

Dataset	Statistic	p-value	Result
Imputed Cryotherapy	20.59928	0	H is accepted
Imputed Immunotherapy	19.05097	0	H is accepted

Table 10
Shapiro-Wilk test (significance level of 0.10) on imputed cryotherapy.

Dataset Feature	Statistic	p-value	Result
sex	0.46873	0.00000	H is rejected
age	0.85001	0.00000	H is rejected
Time	0.63570	0.00000	H is rejected
Number_of_Warts	0.92219	0.00005	H is rejected
Type	0.90677	0.00001	H is rejected
Area	0.63480	0.00000	H is rejected
Result_of_Treatment	0.66495	0.00000	H is rejected

Table 11
Shapiro-Wilk test (significance level of 0.10) on imputed Immunotherapy.

Dataset	Statistic	p-value	Result
sex	0.64832	0.00000	H is rejected
age	0.54736	0.00000	H is rejected
Time	0.92937	0.00011	H is rejected
Number_of_Warts	0.63354	0.00000	H is rejected
Type	0.95623	0.00414	H is rejected
Area	0.91939	0.00003	H is rejected
Result_of_Treatment	0.73803	0.00000	H is rejected

Porwik, 2021). The suggested technique outperforms all other techniques.

Multiple hypothesis testing with Bonferroni correction using a two-sample *t*-test is implemented with the table one package (Pollard et al., 2018). A comparison between the proposed and other imputation techniques on the Cryotherapy and Immunotherapy datasets is shown in Table 7 and Table 8, respectively. The results show that the hypothesis is accepted, and the results are close for all approaches. For the cryotherapy dataset, MICE and Median imputation approaches achieve the best p-value on the number of warts, while k-NN achieves the best with the area. For the immunotherapy dataset, k-NN achieves the best p-values on age and time, while the proposed imputation method achieves the best with the number of warts and types.

We have used the Levene test on both datasets, the imputed cryotherapy and the imputed Immunotherapy, with a significance level of 0.1 where the null hypothesis (H) is accepted if the p-value is lower than 0.1. Table 9 shows the Leven statistics result. The results show that the imputation didn't cause a negative impact on the homoscedasticity of the data.

Also, Shapiro-Wilk statistical test is done for both datasets. The hypothesis (H) is accepted if the p-value is above 0.1, which is the significance level used. Table 10 shows the test result on the imputed Cryotherapy dataset, while Table 11 shows it on the imputed Immunotherapy dataset. The p-value results show that the imputed values didn't cause a negative impact on the dataset normality.

To investigate the impact of the dimensionality of data and size, we used three datasets with different dimensionality (differs from 4 features to 12 features) and different sizes (differs from 150 to 4898 instances). The used datasets are Iris (C, 2012), wine quality (Cortez et al., 2009), and abalone (Treadgold and Gedeon, 1997). We also tested the execution time on different datasets. We artificially made missing values in the dataset and imputed them using the proposed imputation algorithm. The imputation results are then compared with the original values to get the RMSE. The maximum, minimum, and average values of RMSE are shown in Table 12. The results show that the imputation method has a low RMSE compared to the dataset size. The result of the datasets is shown in Table 11. From the results, we can conclude that the proposed algorithm has efficient execution time even when dealing with large datasets in terms of size or dimensionality. This advantage comes from FKTM scanning only top k values rather than scanning the whole dataset.

6. Conclusion

In this paper, we provide Top-k Case Matching, a straightforward, efficient, and reliable method for imputing missing values in different datasets. The fundamental idea behind the method is to locate the k situations in previous data that are the most comparable to the current circumstance and extract missing values from the data during these periods. Our Top-k Case Matching approach is based on fuzzy theory. The results of the experimental investigation were presented. The Top-k Case Matching approach produces good results in terms of accuracy, error rate, sensitivity, specificity, and precision and may even outperform the original dataset. Three classifiers were utilized to compare the performance metrics of the imputed and original datasets. The suggested technique was tested and compared to earlier work on two well-known datasets. Our

Table 12
Imputation Results on Iris, Wine Quality, and Abalone datasets.

Dataset	RMSE(MIN)	RMSE(MAX)	RMSE(AVG)	exe_time	Size*dimension	Noise Ratio
Iris	1.6903	4.66	2.90464	0.1092	150*4	0.05
Wine Quality	0.01	204.4328	25.45980833	1.5288	4898*12	0.05
Abalone	0.285	19.8089	3.461925	1.4193	4177*8	0.05

findings reveal that the recommended FKTM outperforms the original FKTM since imputed values in missing datasets cause imputed datasets to be classified in the same way as original datasets, with minor differences in performance metrics. We also tested our imputation strategy using two statistical methods, the Shapiro-Wilk and Levene, where the proposed methods achieved good results. Also, the influence of the size of the data in terms of the number of samples and data dimensionality is explored on the RMSE and execution time, and the system achieved promising results. Our future research direction is to optimize the complexity of the proposed technique and increase its imputation accuracy.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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